

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

ASSESSMENT TOOL 2 – Coding Golf Competition

Course Code:	CSA03	Course Title:	Data Structures
CO Assessed:	CO4 –Articulation (BL3)	Assessment:	Assessment Tool 2 – Coding Golf Competition
Weightage:	20% of CIA	Total Marks:	50
CO3 Statement:	Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data		
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Section / Batch:	B-TECH(IT)	Date:	12-08-2026

Code of Conduct

I **T.VAISHNAVI(192421370)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 Coding challenge Total: 50 Marks.
- Evaluation of Answer Quality
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Hashing - Priority Queue & Heaps-	Hashing, Heap	1,2	20
Sorting Algorithm	Insertion sort, Heap sort	3,4	20
Parallel Sorting algorithms-Time complexity.	Merge Sort, Quick sort	5	10
GRAND TOTAL:		50Marks	

Annexure A – Coding Golf Competition

Q1) Linear Probing – Basic

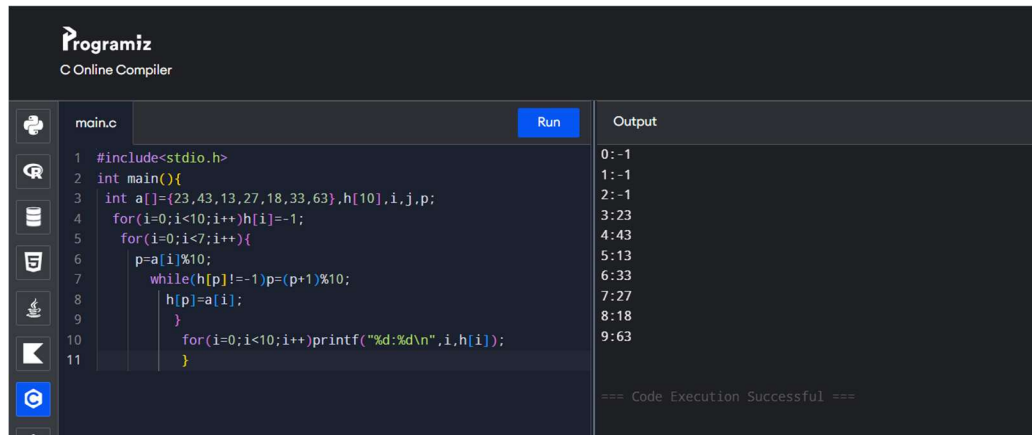
Given the keys:

23, 43, 13, 27, 18, 33, 63

and hash function:

Write a program to insert all keys into a hash table using linear probing. Display the final hash table.

Golf condition: Minimum number of lines of correct code.



```
1 #include<stdio.h>
2 int main(){
3     int a[]={23,43,13,27,18,33,63},h[10],i,j,p;
4     for(i=0;i<10;i++)h[i]=-1;
5     for(i=0;i<7;i++){
6         p=a[i]%10;
7         while(h[p]!=-1)p=(p+1)%10;
8         h[p]=a[i];
9         for(i=0;i<10;i++)printf("%d:%d\n",i,h[i]);
10    }
11 }
```

Output

```
0:-1
1:-1
2:-1
3:23
4:43
5:13
6:33
7:27
8:18
9:63

=== Code Execution Successful ===
```

Q2) Separate Chaining

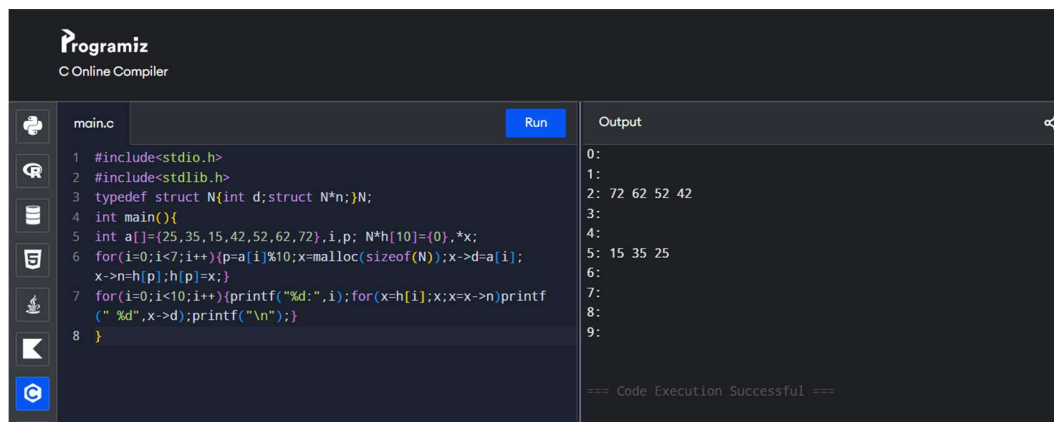
Implement a hash table of size 10 using separate chaining for:

25, 35, 15, 42, 52, 62, 72

using:

Display the resulting chains.

Golf condition: Produce the shortest correct implementation.



```
1 #include<stdio.h>
2 #include<stdlib.h>
3 typedef struct N{int d;struct N**N;}N;
4 int main(){
5     int a[]={25,35,15,42,52,62,72},i,p; N*h[10]={0},*x;
6     for(i=0;i<7;i++){p=a[i]%10;x=malloc(sizeof(N));x->d=a[i];
7     x->n=h[p];h[p]=x;}
8     for(i=0;i<10;i++){printf("%d:",i);for(x=h[i];x=x->n)printf
9     (" %d",x->d),printf("\n");}
```

Output

```
0:
1:
2: 72 62 52 42
3:
4:
5: 15 35 25
6:
7:
8:
9:

=== Code Execution Successful ===
```

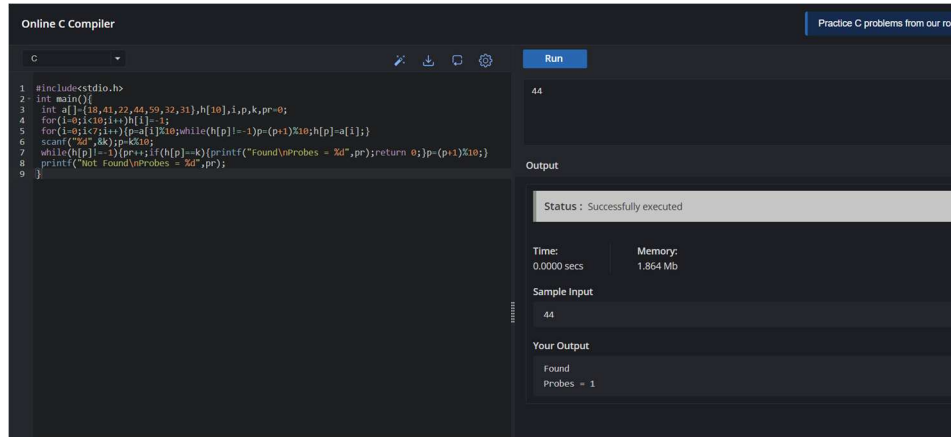
Q3) Search Using Hashing

Given:

18, 41, 22, 44, 59, 32, 31

construct a hash table using linear probing and write a program to search for a given key.

Challenge: The program should report the number of probes required.



```
1 #include <stdio.h>
2 int main()
3 {
4     int a[] = {18, 41, 22, 44, 59, 32, 31}, h[10], i, p, k, pr = 0;
5     for(i = 0; i < 10; i++) h[i] = -1;
6     for(i = 0; i < 7; i++) {p = a[i] % 10; while(h[p] != -1) p = (p + 1) % 10; h[p] = a[i];}
7     scanf("%d", &k); p = k % 10;
8     while(h[p] != -1) {pr++; if(h[p] == k) {printf("Found\nProbes = %d", pr); return 0;} p = (p + 1) % 10;}
9     printf("Not found\nProbes = %d", pr);
10 }
```

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.864 Mb

Sample Input

44

Your Output

Found
Probes = 1

Q4) Collision Counter

Write a program that inserts N keys into a hash table using linear probing and counts the total number of collisions.

Example:

Input:

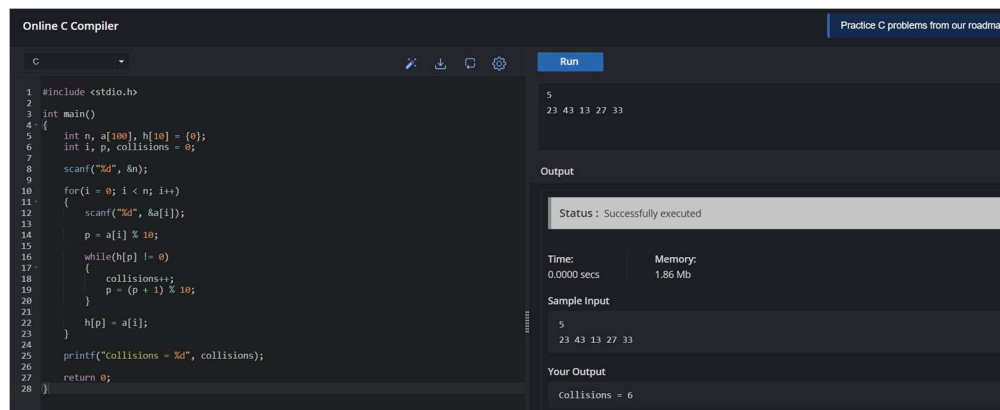
5

23 43 13 27 33

Output:

Collisions = ?

Golf: Shortest correct code wins.



```
1 #include <stdio.h>
2 int main()
3 {
4     int n, a[100], h[10] = {0};
5     int i, p, collisions = 0;
6     scanf("%d", &n);
7     for(i = 0; i < n; i++)
8     {
9         scanf("%d", &a[i]);
10        p = a[i] % 10;
11        while(h[p] != 0)
12        {
13            collisions++;
14            p = (p + 1) % 10;
15        }
16        h[p] = a[i];
17    }
18    printf("Collisions = %d", collisions);
19    return 0;
20 }
```

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.86 Mb

Sample Input

5
23 43 13 27 33

Your Output

Collisions = 6

Q5) Q5. Kth Largest Element

Given:

15 7 20 10 3 25 18

and K = 3, find the 3rd largest element using a heap.

Expected output:

18

Golf condition: Do not use a built-in sorting function.

```

1 #include<stdio.h>
2 void heap(int a[],int n,int i){
3     int m=i*2,i>=n?i:2,t;
4     if(1<n&&a[1]>a[m])m=1;
5     if(r<n&&a[r]>a[m])m=r;
6     if(m!=i){t=a[i];a[i]=a[m];a[m]=t;heap(a,n,m);}
7 }
8 int main(){
9     int a[]={15,7,20,10,3,25,18},n=7,k=3,i,t;
10    for(i=n/2-1;i>=0;i--)heap(a,n,i);
11    for(i=n-1;i>=n-k+1;i--){t=a[0];a[0]=a[i];a[i]=t;heap(a,i,0);}
12    printf("%d",a[0]);
13 }
  
```

Run

15 7 20 10 3 25 18
K = 3

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.708 Mb

Sample Input

15 7 20 10 3 25 18
K = 3

Your Output

18

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct output	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Algorithm Implementation	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Code optimization	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Time complexity	20%	Demonstrates strong critical thinking with clear interpretation and	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation

		reasoning			
Code readability	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____